

Sri Lanka Institute Of Information Technology



IT3021 – Data Warehousing & Business Intelligence Assignment Report 2

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STEP 1 — Data Source

The **FoodDelivery_DW**, which was created in Assignment 1 using SQL Server, is the data warehouse utilized for this assignment.

The warehouse uses a star schema design, with several dimension tables storing descriptive attributes and a central fact table holding transactional data.

Tables

- Fact Table - Fact_Orders
- Dimension Tables
 - Dim_Customer
 - Dim_Restaurant
 - Dim_Payment
 - Dim_Date

The Fact_Orders table stores transactional order data including order value and references to dimensions

Dimension tables provide descriptive information about customers, restaurants, payment methods, and dates.

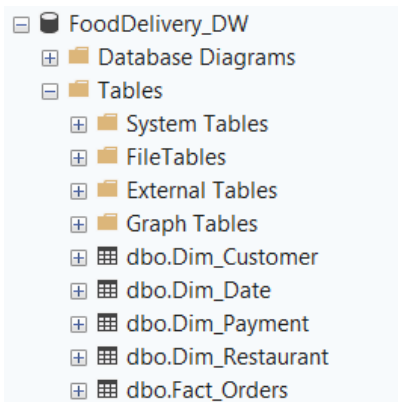


Figure 2: Food Delivery Data Warehouse Tables

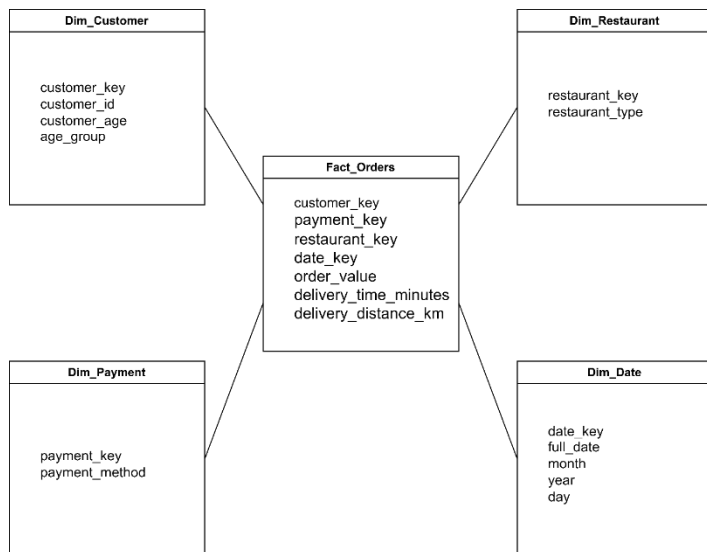


Figure 1: Star Schema of Food Delivery Data Warehouse

STEP 2 — SSAS Cube Implementation

1. SQL Server Data Tools (SSDT) was used to create a new Analysis Services Multidimensional Project.
2. A data source connection was created to connect the SSAS project with the FoodDelivery_DW database using Windows Authentication.

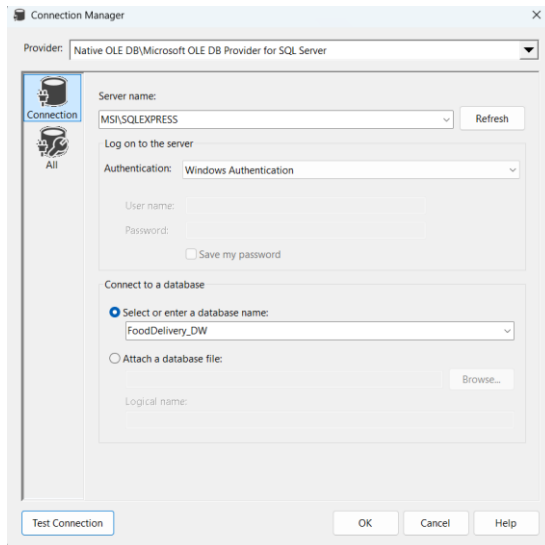


Figure 3: Data Source Configuration

3. A Data Source View (DSV) was created by importing the fact and dimension tables from the data warehouse.

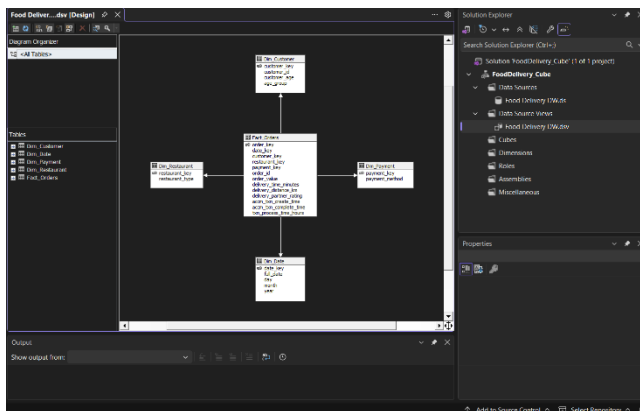


Figure 4: Data Source View Showing Relationships

4. The cube was created using the Cube Wizard by selecting Fact_Orders as the measure group
5. Add Dimensions

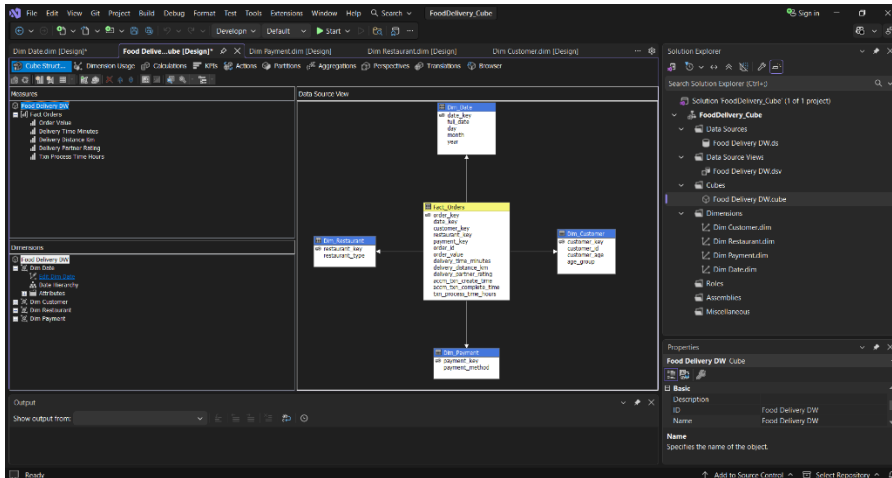


Figure 5: Cube Structure Showing Measures and Dimensions

6. A date hierarchy was created in the **Dim_Date** dimension to enable hierarchical navigation

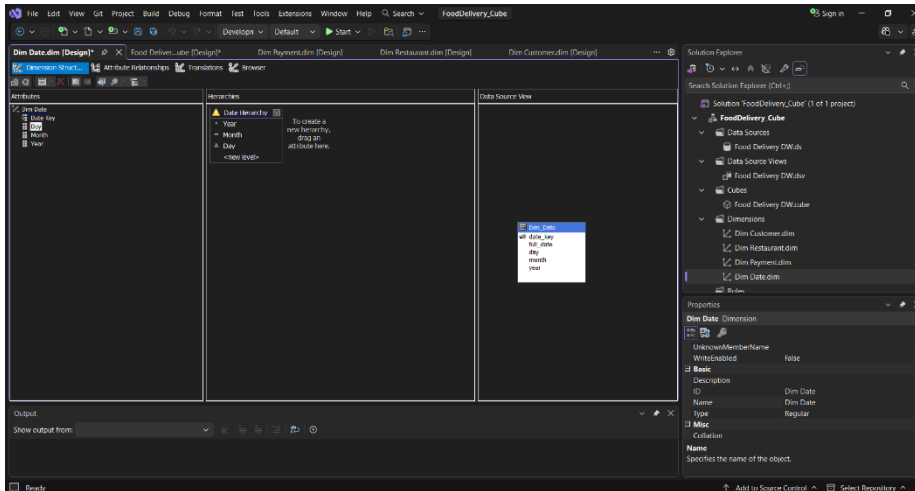


Figure 6:Hierarchy creation screen

7. The cube was deployed to the Analysis Services server and processed using the Process Full option

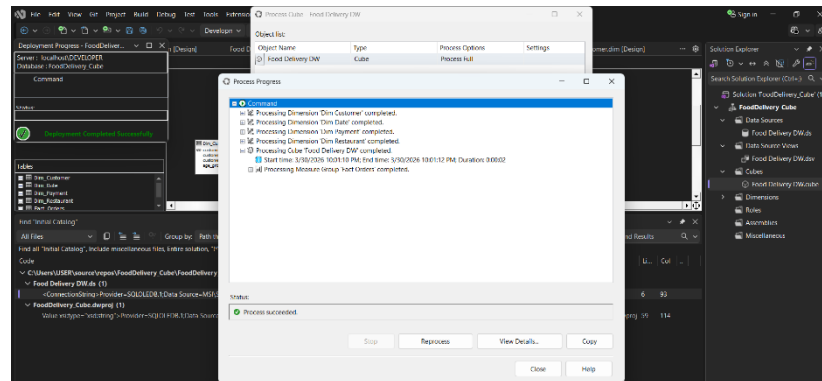


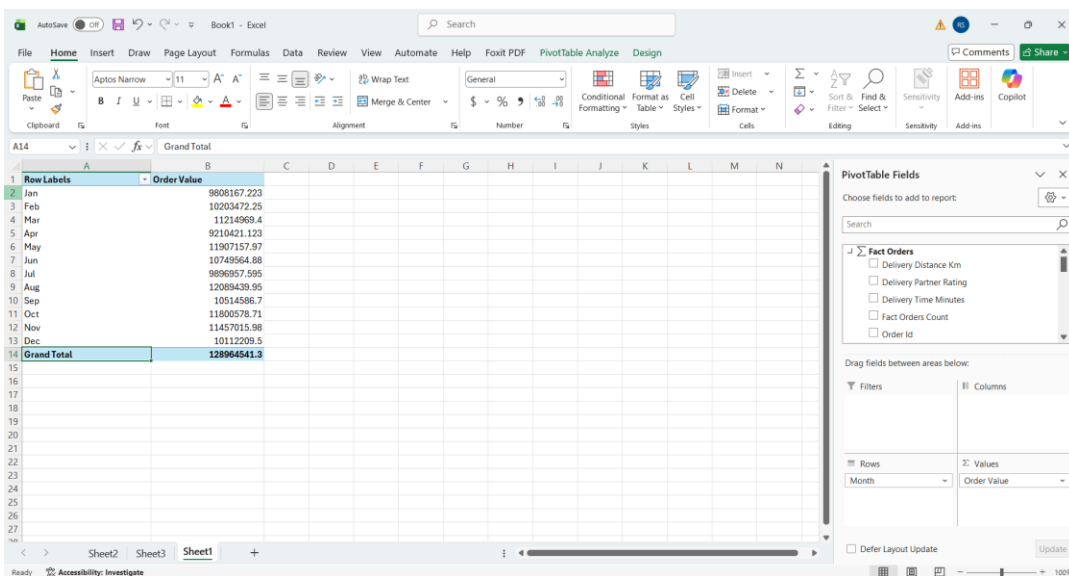
Figure 7:Successful Cube Deployment

STEP 3 — OLAP Operations

1. Connected Excel to SSAS Cube – Used Excel's Get Data feature to establish a connection to the deployed Hotel Reservation SSAS cube using the SQL Server Analysis Services server
2. Created PivotTable – Generated a PivotTable report in Excel linked to the cube to perform multidimensional analysis

Roll – Up

Aggregates data from a lower level to a higher level



Drill-Down

Allows users to navigate from summarized data to detailed data

	A	B	C
1	Row Labels	Order Value	
2	2024	128964541.3	
3	Grand Total	128964541.3	
4			
5			

Figure 8: Year

Row Labels	Order Value
2024	
Jan	9808167.223
Feb	10203472.25
Mar	11214969.4
Apr	9210421.123
May	11907157.97
Jun	10749564.88
Jul	9896957.595
Aug	12089439.95
Sep	10514586.7
Oct	11800578.71
Nov	11457015.98
Dec	10112209.5
Grand Total	128964541.3

Figure 10: Month

Row Labels	Order Value
2024	
Jan	
1	91451.42014
2	287814.1226
3	289861.7036
4	210928.8633
5	350698.5799
6	379296.6226
7	72016.8009
8	353480.2385
9	258321.4201
10	437227.5208
11	153994.7407
12	363026.2452
13	475670.5163
14	411613.1575
15	251463.8425
16	340156.5428
17	216027.4663
18	279455.0787
19	448524.3611
20	204219.2823
21	424907.257
22	594123.8077
23	233099.2593
24	527508.257
25	412261.405
26	166101.1389
27	241681.7823
28	459716.1091
29	427969.3842
30	338865.2385

Figure 9: Date

Slice

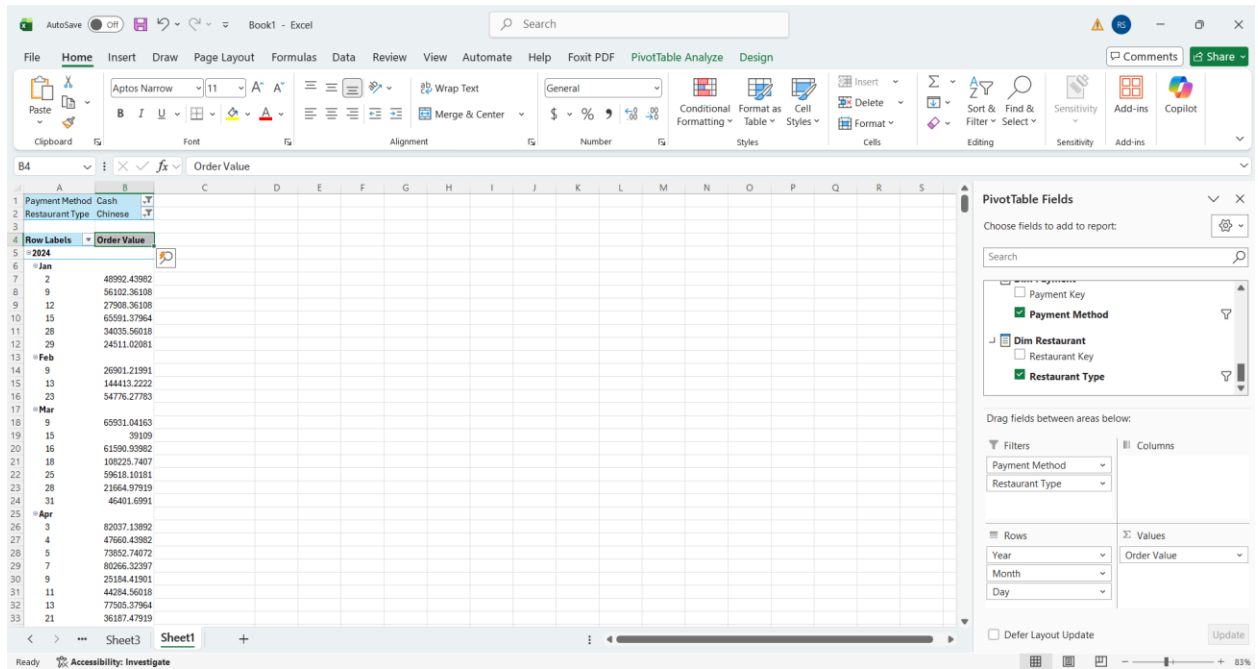
Filters the cube using one dimension

The screenshot displays the Microsoft Excel interface with a PivotTable and the PivotTable Fields task pane. The PivotTable is filtered by the year 2024 and shows the Order Value for each month. The PivotTable Fields task pane is open on the right, showing the 'Payment Method' field selected as the filter. The 'Dim Payment' field is also selected, and the 'Payment Method' field is listed in the 'Columns' area. The 'Rows' area is empty, and the 'Values' area shows 'Order Value'.

Row Labels	Order Value
2024	
Jan	
1	91451.42014
2	287814.1226
3	289861.7036
4	210928.8633
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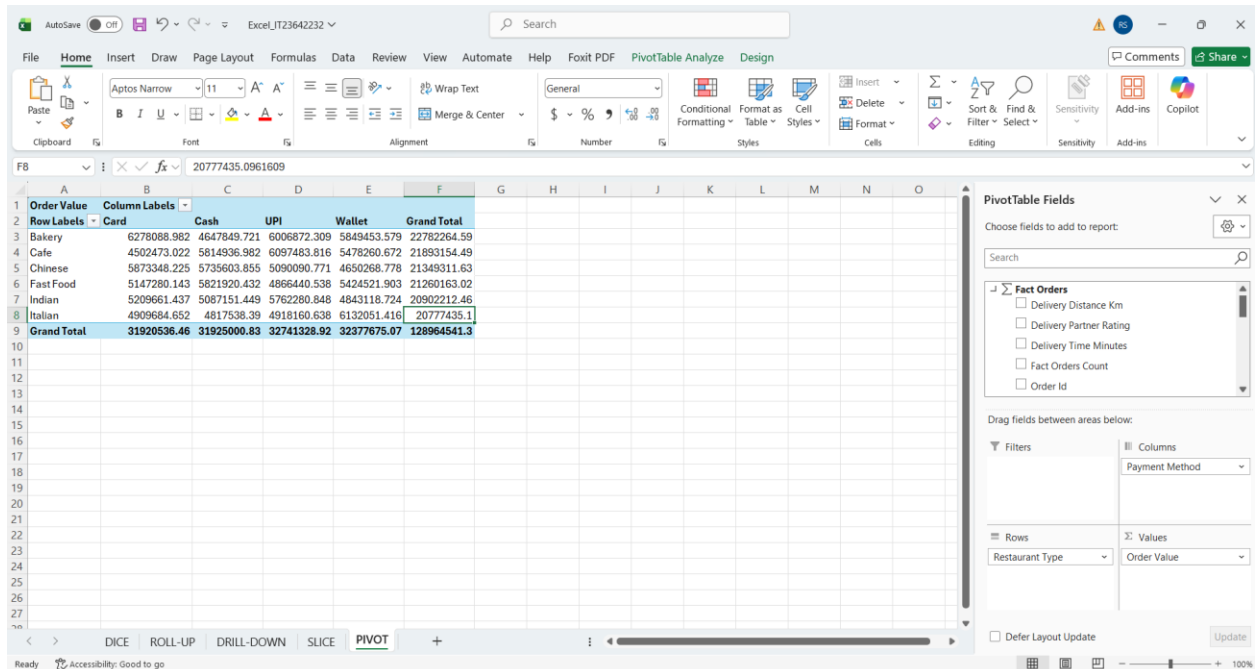
Dice

Filters the cube using multiple dimensions



Pivot

Rotates the cube to change the data perspective



STEP 4 — Power BI Reports

Power BI Report Development

Power BI was used to develop interactive business intelligence reports based on the FoodDelivery_DW data warehouse. The reports were designed to provide insights into customer behavior, restaurant performance, payment methods, and delivery metrics.

The reports include matrix visualization, slicers with cascading filters, drill-down capabilities, and drill-through analysis

Data Preparation & Modeling

After loading the data, relationships between fact and dimension tables were automatically detected and verified. The star schema structure was maintained, where the Fact_Orders table is connected to dimension tables

Fact table:

- Fact_Orders

Dimension tables:

- Dim_Customer
- Dim_Date
- Dim_Restaurant
- Dim_Payment

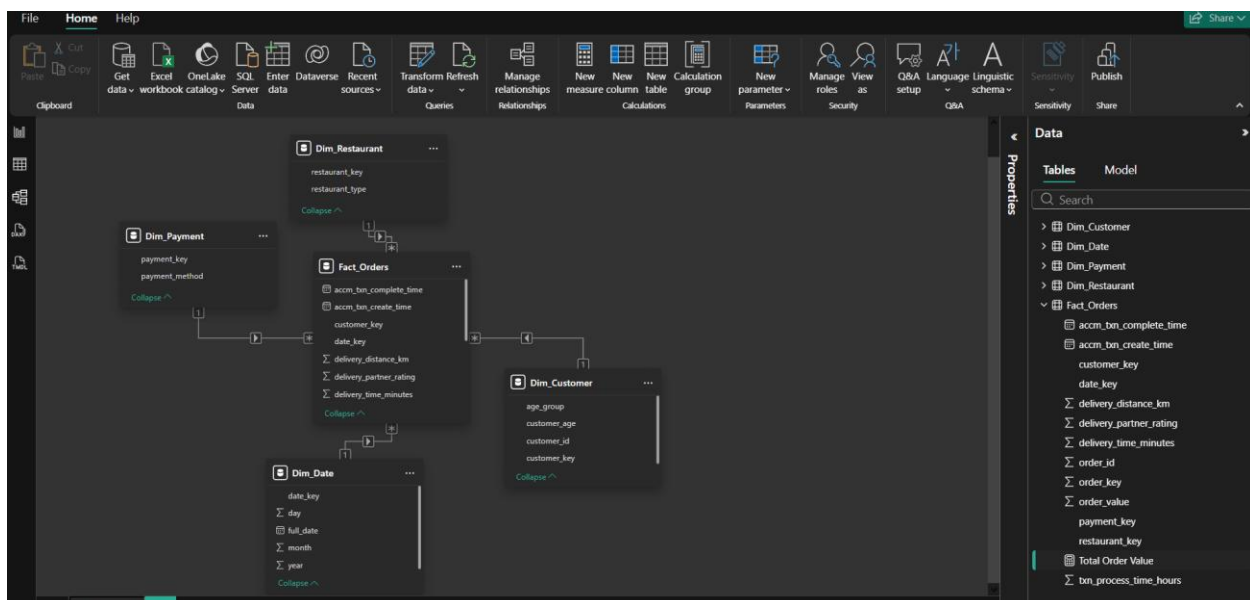


Figure 11: Model view

The SSAS cube already contained predefined **hierarchies and relationships**, therefore no additional relationships were created in Power BI Desktop

The following hierarchy was used:

Date Hierarchy: Year → Month → Day

DAX Calculations

DAX (Data Analysis Expressions) was used to create calculated measures

```
1 Total Order Value =  
2 SUM(Fact_Orders[order_value])
```

Creating Visuals

Power BI link - <https://app.powerbi.com/groups/me/reports/e3a5e27b-f755-4942-8d43-5b139c8bb11b/ab8a27e2b34233d836bb?experience=power-bi>

REPORT 1 — MATRIX VISUAL

A Matrix visual was created to display aggregated order values across different restaurant types and payment methods.

This report helps identify which payment methods are most commonly used across different restaurant categories

- Rows : restaurant_type
- Columns : payment_method
- Values : order_value

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a data table with the following structure:

restaurant_type	Card	Cash	UPI	Wallet	Total
Bakery	6,278,088.98	4,647,849.72	6,006,872.31	5,849,453.58	22,782,264.59
Cafe	4,502,473.02	5,814,936.98	6,097,483.82	5,478,260.67	21,893,154.49
Chinese	5,873,348.22	5,735,603.85	5,090,090.77	4,650,268.78	21,349,311.63
Fast Food	5,147,280.14	5,821,920.43	4,866,440.54	5,424,521.90	21,260,163.02
Indian	5,209,661.44	5,087,151.45	5,762,280.85	4,843,118.72	20,902,212.46
Italian	4,909,684.65	4,817,538.39	4,918,160.64	6,132,051.42	20,777,435.10
Total	31,920,536.46	31,925,000.83	32,741,328.92	32,377,675.07	128,964,541.28

To the right of the table is the 'Filters' pane, which is currently empty. Below the table is a 'Visualizations' pane with a 'Build visual' section and a 'Data' pane showing a list of data fields including Dim_Customer, Dim_Date, Dim_Payment, Dim_Restaurant, and Fact_Orders.

REPORT 2 — SLICERS (CASCADING FILTERS)

Multiple slicers were created to allow dynamic filtering. Cascading filtering was implemented so that selecting a value in one slicer affects the available options in another.

This report enables users to interactively explore data and analyze trends based on selected filters

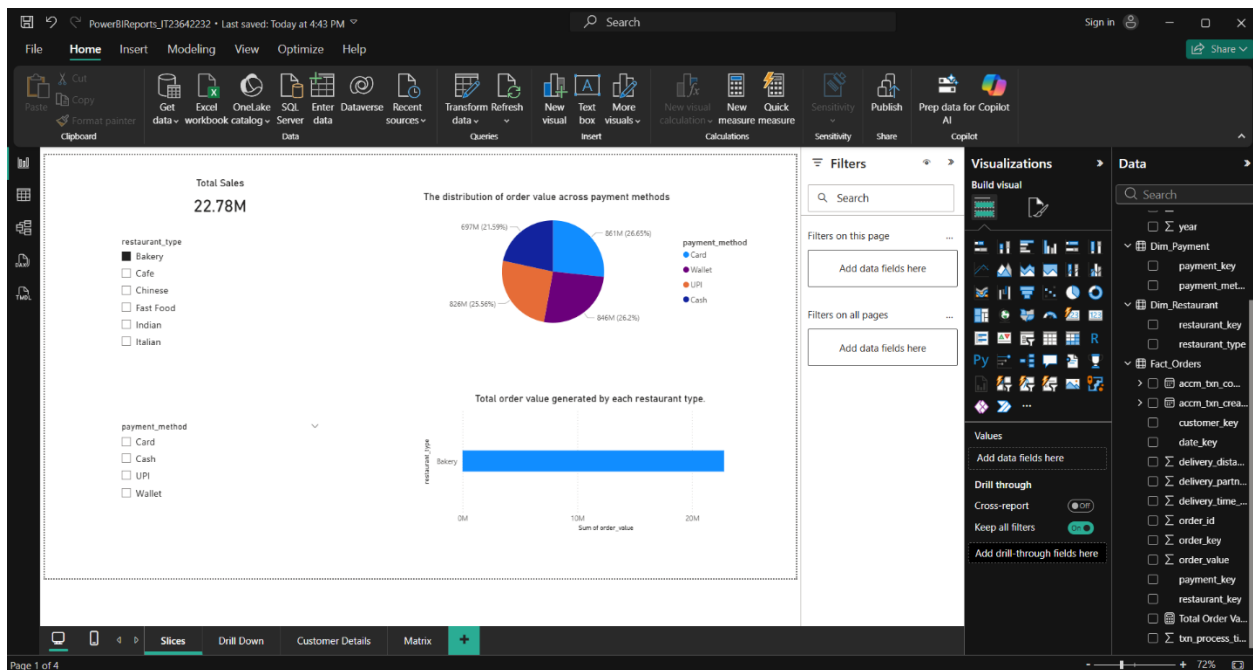


Figure 13: Restaurant type

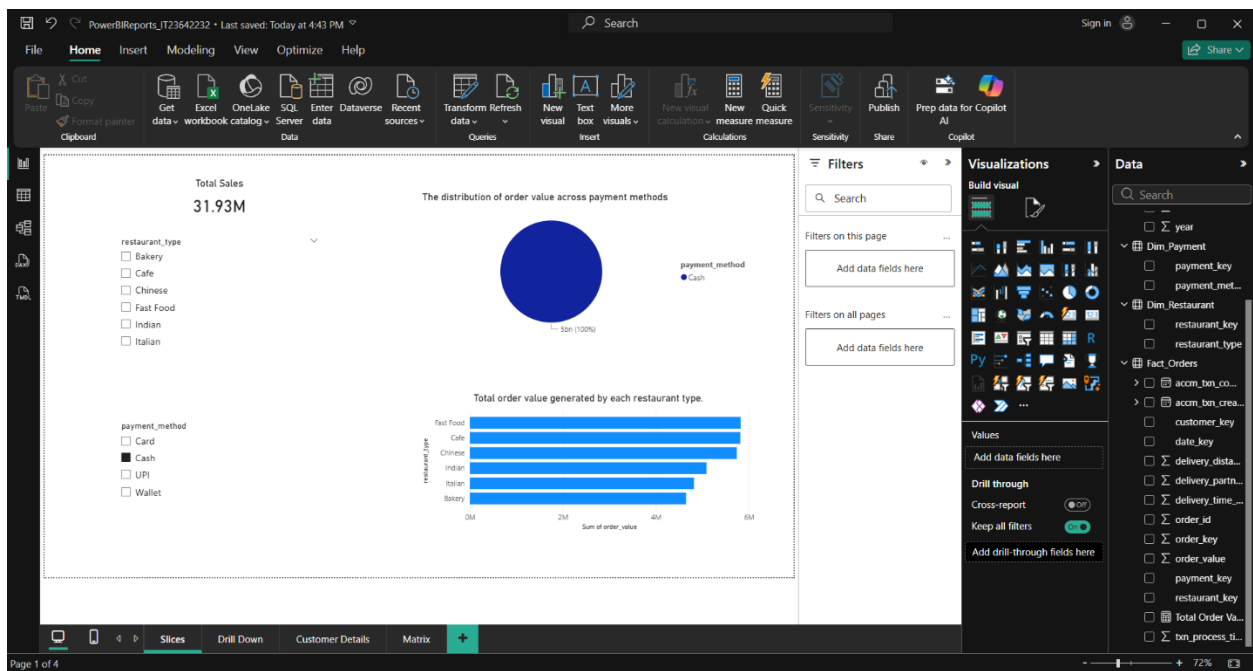


Figure 12: Payment method

REPORT 3 — DRILL-DOWN REPORT

A drill-down feature was implemented using the date hierarchy, allowing users to explore data from year to month and day levels

This feature allows detailed time-based analysis of orders and helps identify seasonal trends



Sum of order value- Drill Down



Sum of order value- Drill Down



Sum of order value- Drill Down

Figure 14:Drill down by year



Figure 17:Drill down by quarter



Figure 16::Drill down by date

REPORT 4 — DRILL-THROUGH REPORT

A drill-through report was created to allow users to navigate to a detailed view of data by right-clicking on a visual

This enables users to analyze detailed information related to a selected data point

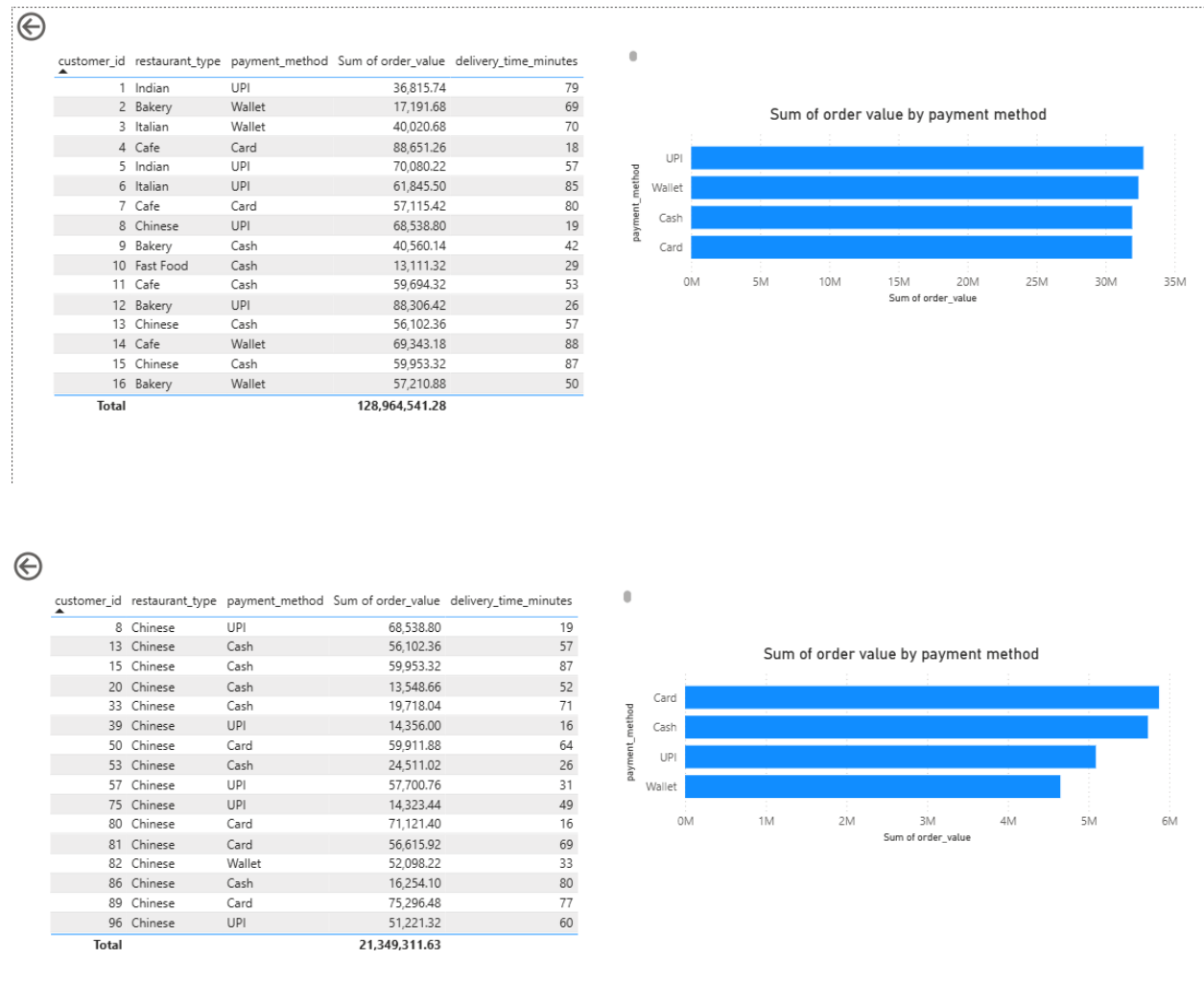


Figure 18: Drill through by restaurant type (Chinese)

Summary

This assignment showed how to use Microsoft tools to fully implement a data warehousing and business intelligence solutions. FoodDelivery_DW, a data warehouse created in Assignment 1, was effectively utilized as the basis for additional analytical processing.

Fact tables, dimensions, and measures were defined in order to create and implement an SSAS multidimensional cube. A hierarchy (Year → Month → Day) was developed to facilitate time-based analysis, and relationships between dimensions and the fact table were appropriately established. SQL Server Analysis Services was used to successfully deploy and process the cube.

Microsoft Excel was used to connect to the cube and demonstrate OLAP operations. Multi-dimensional data analysis was made possible using pivot tables for important analytical operations like roll-up, drill-down, slice, dice, and pivot.

Additionally, Power BI was used to create insightful and interactive reports. These included drill-down visualizations, drill-through reports, slicers with cascading filters, and a matrix report. To improve report interactivity and analytical capabilities, DAX measures, relationships, and data modelling were employed.

All things considered, this assignment gave me practical experience creating a comprehensive BI solution by combining data warehousing, OLAP analysis, and contemporary data visualization tools. By providing insightful information, the implemented solution facilitates effective data analysis and improves decision-making.